

## QUESTION ONE

(a) Write the quadratic equation in the form  $ax^2 + bx + c = 0$  that has solutions

$$x = \frac{2}{3} \text{ and } x = \frac{-3}{5}$$

(b) Factorise completely  $15ap + 6aq - 5bp - 2bq$ .

(c) Solve  $\frac{12x^2 - x - 6}{3x^2 - 16x - 12} = -2$ .

(d) Auckland is New Zealand's largest city. A local Council report in 2022 shows its population to be approximately 1.74 million people. Current projections show that the population for Auckland is estimated to grow at a rate of 1.12% each year.

(i) By using the model  $P(t) = Ab^t$ , calculate what the estimated population will be in 2035. [ $P(t)$  is the population for Auckland after  $t$  years].

(ii) Auckland City Councillors are extremely concerned that the capital expenditure for the infrastructure of basic services such as water, wastewater, roading, and communication to meet this future population growth.

They have determined that they can sustain the services provided Auckland's population does not exceed 3 million people by the year 2100.

Determine the maximum rate of population growth..

(e) Solve for  $x$  in terms of  $k$   $\frac{\log x}{\log(kx)} = 2$

*Please do NOT write on this paper 😊*

## QUESTION TWO

(a) Simplify this expression, leaving your answer with positive exponents only.

$$\frac{4a^{\frac{5}{4}}}{\sqrt[3]{64a^{12}}}$$

(b) The plan for a new park has an elliptical garden which has the equation:

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

A drain across the park is modelled by the equation:

$$3x + 5y = k$$

Determine the range of values for  $k$  that will prevent the drain crossing the garden.

Show all working with correct mathematical statements.

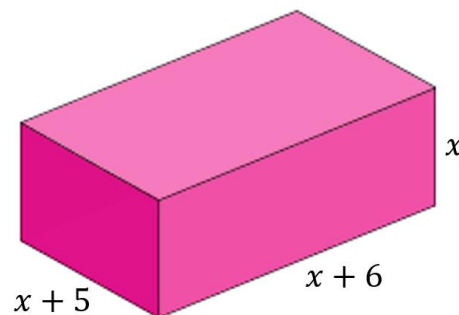
(c) Write  $\frac{12}{x^2-16} + \frac{3}{2x+8}$  as a single fraction in its simplest form.

(d) Given  $16^m = 2^n$  write an expression for  $n$  in terms of  $m$ .

(e) A cuboid made of playdough has the dimensions as shown.  
The cuboid is remoulded into a cube using the same quantity of playdough.

The cube's height is twice the height of the cuboid.

Using algebraic processes, determine the volume of the cube.  
Measurements are in cm.



### QUESTION THREE

(a) (i) Solve  $\log_x \frac{1}{16} = -2$ .

(ii) Write as a single logarithm in its simplest form  $2 \log xy + \log 6 - \log 3y$

(b) Solve for  $x$ :  $6(\log_5 x)^2 + 9 \log_5 x - 6 = 0$ .  
Solve using algebraic processes.

(c) One root of the equation  $x^2 + 16x + k$  is seven times the other. Find the value of  $k$ .

(d) The sum of cubes formula is an important algebraic identity.  
It is represented by the expression  $a^3 + b^3$ .

In factored form  $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ .

(i) Using the above identity, factorise  $27x^3 + 8$ .

(ii) The equation  $\frac{x^3+8}{x^2-4} = k$  has two distinct solutions.  
Determine the range of possible values for  $k$ .